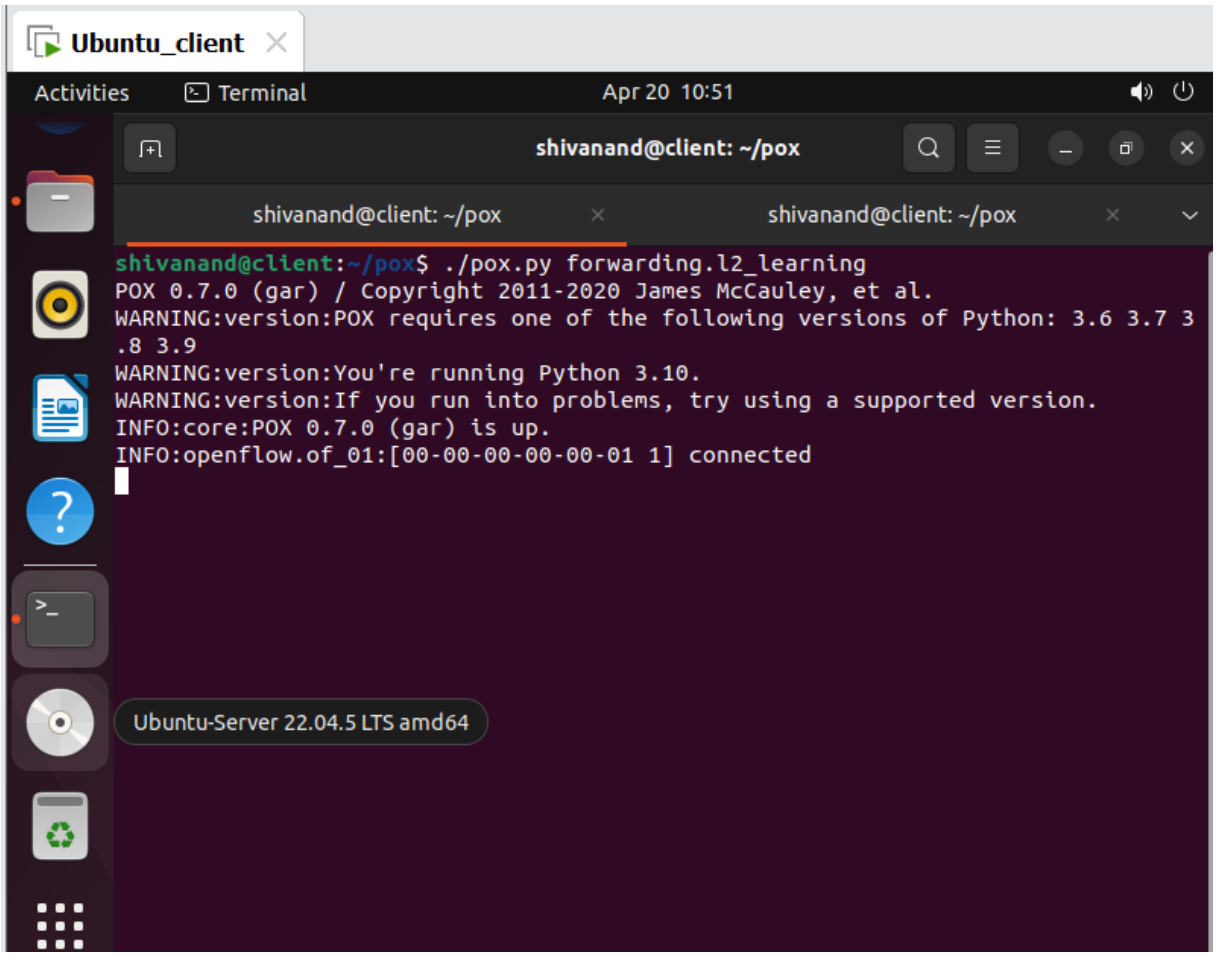
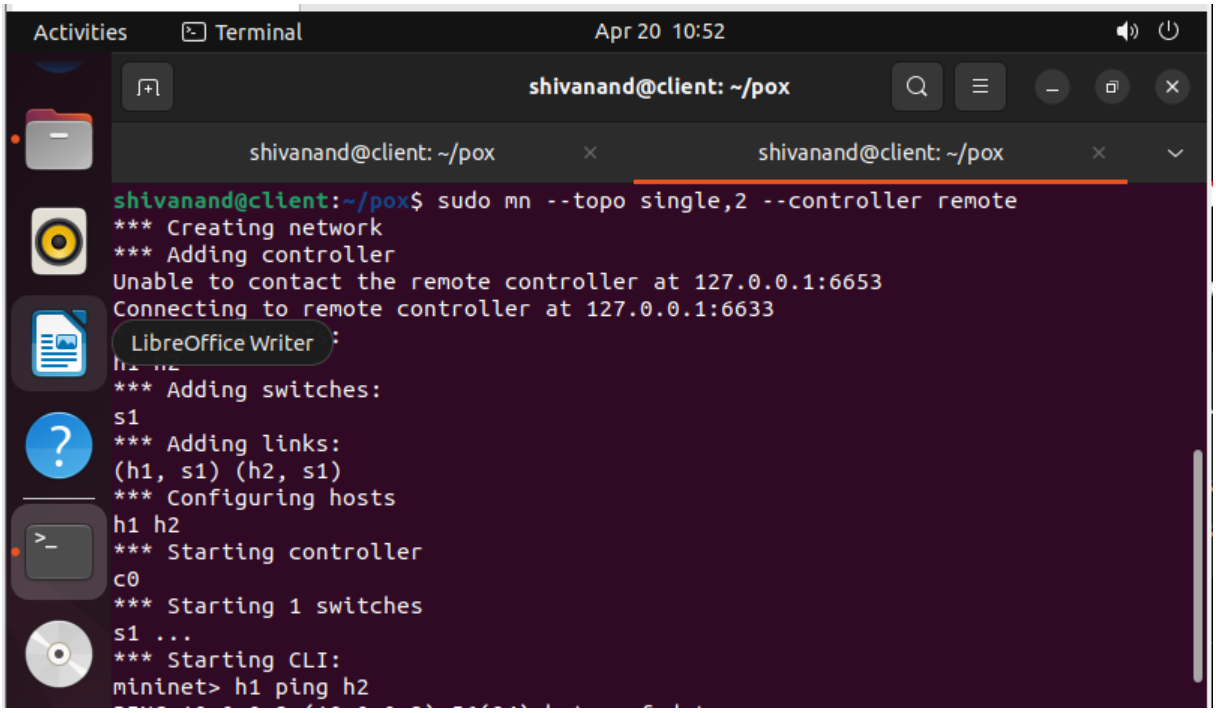
Screenshots / Proof of Execution 1. Mininet Topology

Description: Shows 2 hosts and 1 switch connected to the controller. [Insert

Screenshot Here: Mininet startup output]



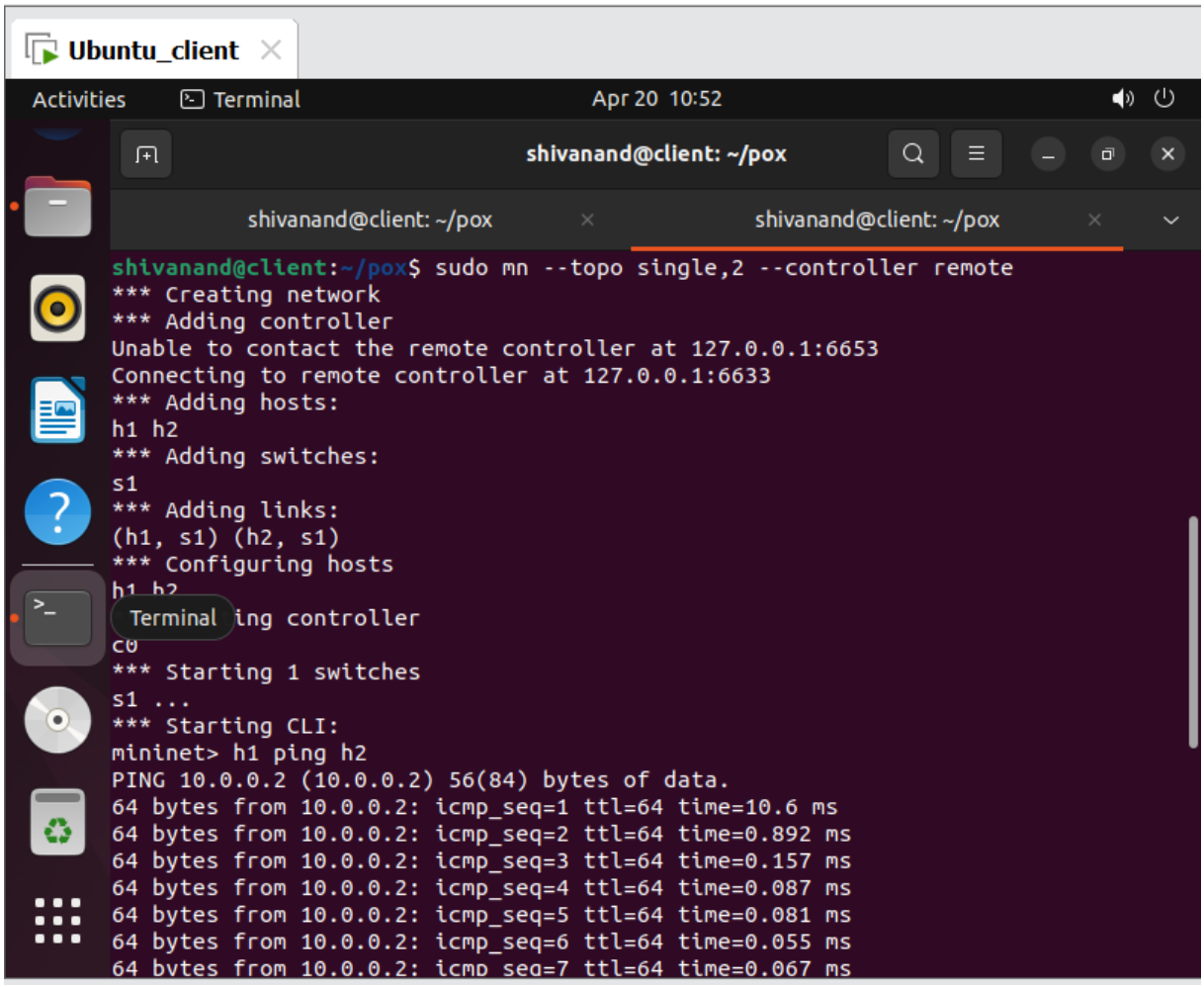


2. Ping Test

Command:
mininet> h1 ping h2

Expected Output:
Ping is successful between h1 and h2.

[Insert Screenshot Here: Ping output]

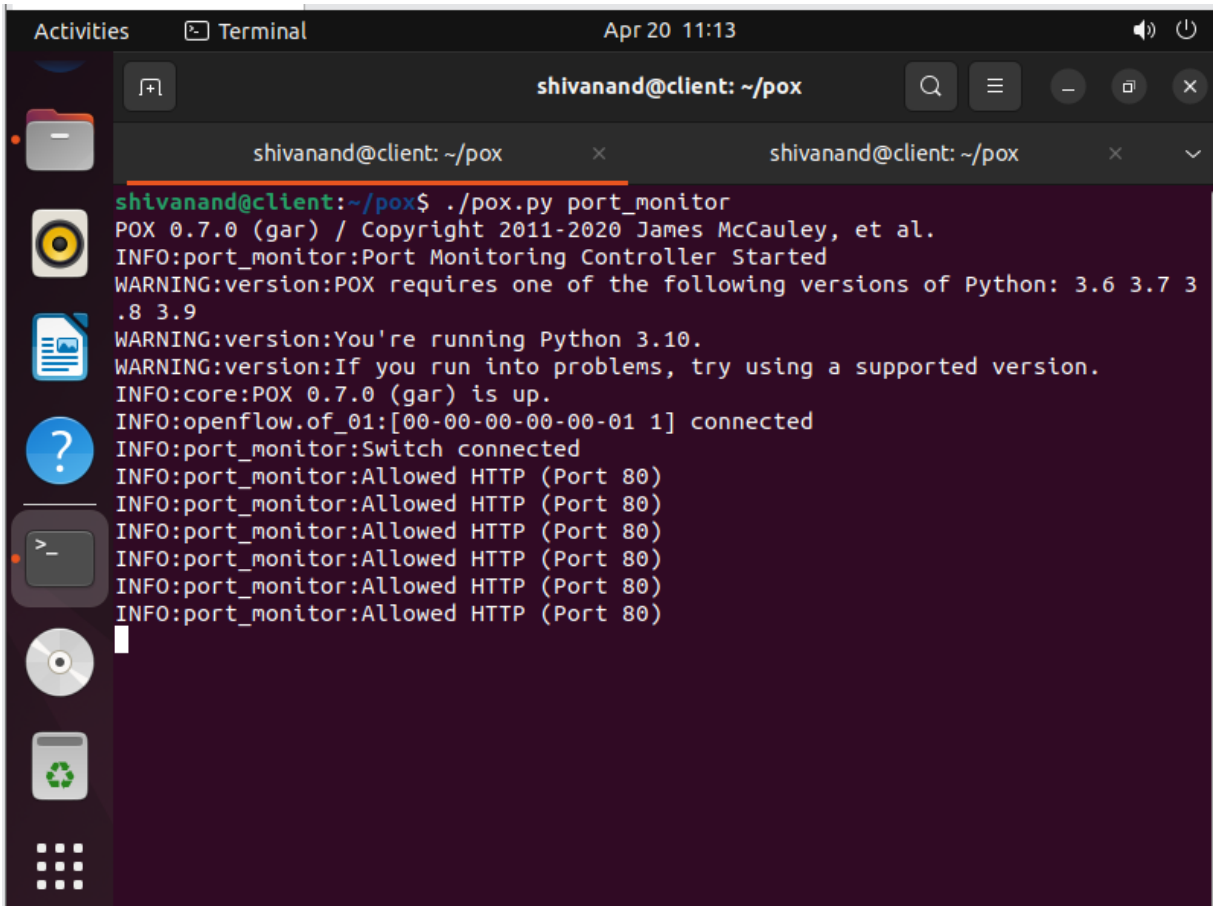


3. HTTP Test (Allowed)

Command:
mininet> h2 python3 -m http.server 80 mininet>
h1 curl http://10.0.0.2

Expected Output:
HTML page is displayed, indicating HTTP traffic is allowed.

[Insert Screenshot Here: HTTP output]


Activities
Terminal
Apr 20 11:13
shivanand@client: ~/pox
shivanand@client: ~/pox
shivanand@client: ~/pox
shivanand@client:~/pox$ ./pox.py port_monitor
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
INFO:port_monitor:Port Monitoring Controller Started
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3
.8 3.9
WARNING:version:You're running Python 3.10.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
INFO:openflow.of_01:[00-00-00-00-00-01 1] connected
INFO:port_monitor:Switch connected
INFO:port_monitor:Allowed HTTP (Port 80)
INFO:port_monitor:Allowed HTTP (Port 80)
INFO:port_monitor:Allowed HTTP (Port 80)
INFO:port_monitor:Allowed HTTP (Port 80)
INFO:port_monitor:Allowed HTTP (Port 80)
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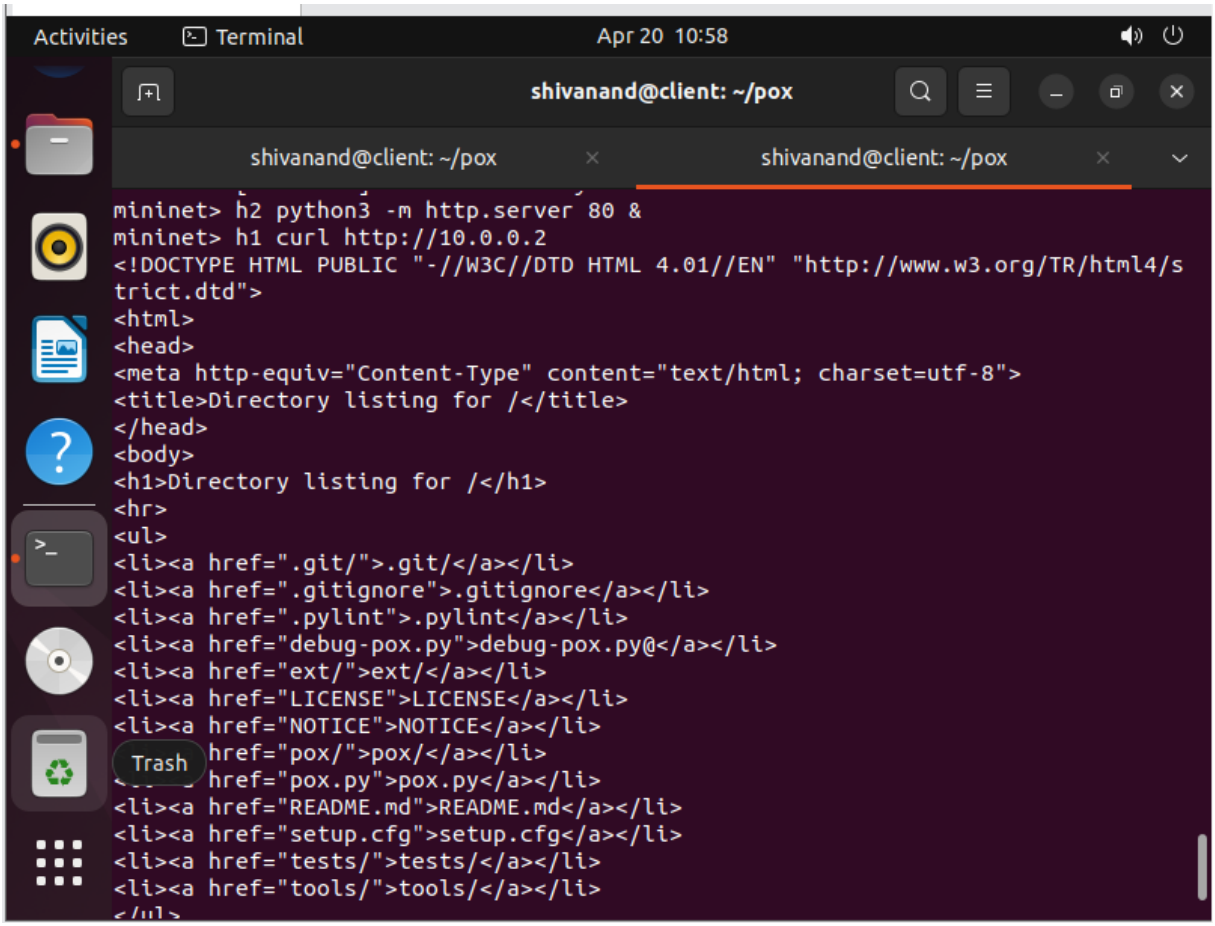
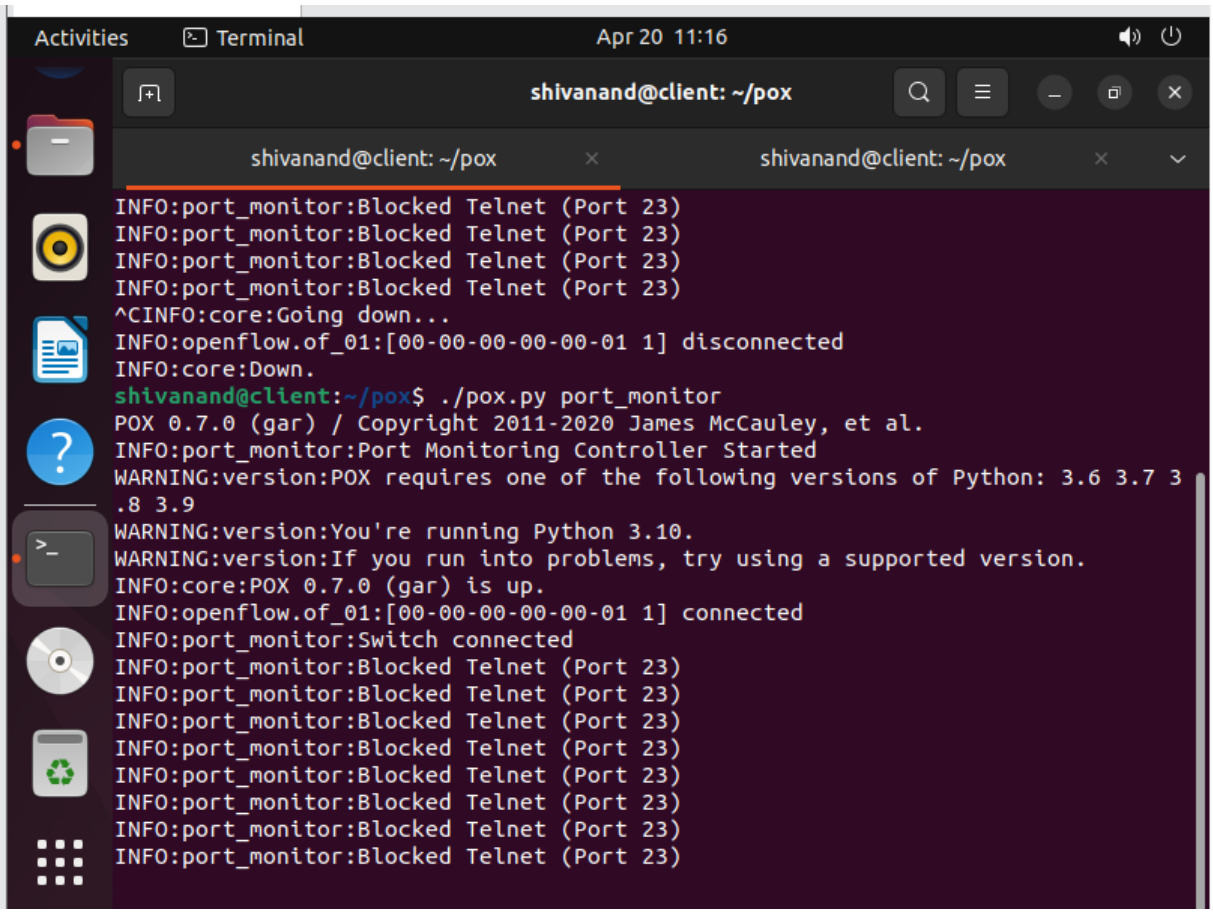


4. Telnet Test (Blocked)

Command:
mininet> h1 telnet 10.0.0.2 23

Expected Output:
Connection fails, indicating Telnet traffic is blocked.

[Insert Screenshot Here: Telnet output]


Activities
Terminal
Apr 20 11:16
shivanand@client: ~/pox
shivanand@client: ~/pox
shivanand@client: ~/pox
INFO:port_monitor:Blocked Telnet (Port 23)
INFO:port_monitor:Blocked Telnet (Port 23)
INFO:port_monitor:Blocked Telnet (Port 23)
INFO:port_monitor:Blocked Telnet (Port 23)
^CINFO:core:Going down...
INFO:openflow.of_01:[00-00-00-00-00-01 1] disconnected
INFO:core:Down.
shivanand@client:~/pox$ ./pox.py port_monitor
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
INFO:port_monitor:Port Monitoring Controller Started
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3
.8 3.9
WARNING:version:You're running Python 3.10.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
INFO:openflow.of_01:[00-00-00-00-00-01 1] connected
INFO:port_monitor:Switch connected
INFO:port_monitor:Blocked Telnet (Port 23)
INFO:port_monitor:Blocked Telnet (Port 23)
INFO:port_monitor:Blocked Telnet (Port 23)
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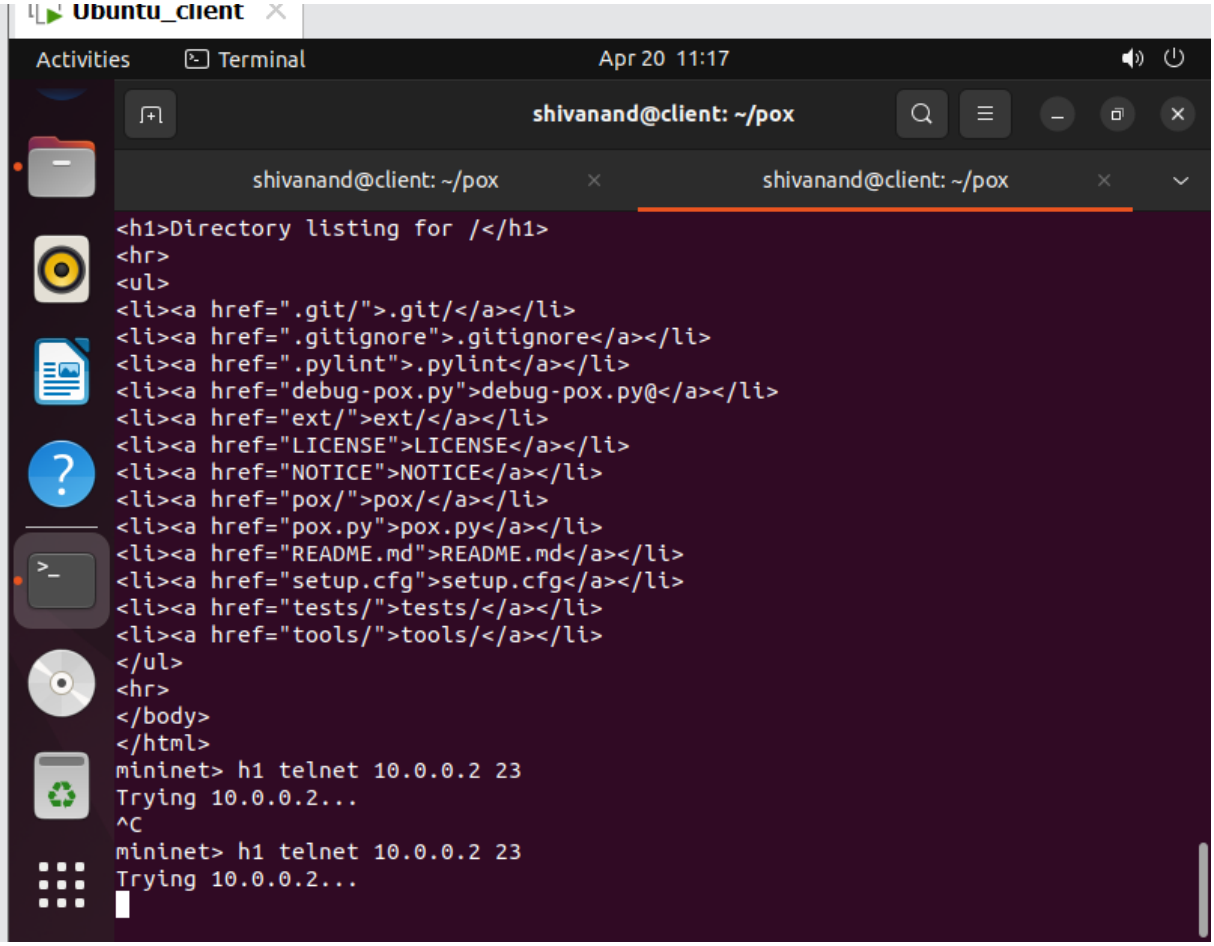
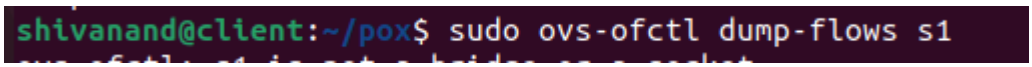


5. Flow Table Verification

Command:
sudo ovs-ofctl dump-flows s1

Expected Output:
Flow entries displayed in the switch.

[Insert Screenshot Here: Flow table]



Explanation :

The flow table shows minimal or no entries because the controller processes packets dynamically using packet_out messages instead of installing static flow rules. This behavior is expected in this implementation."

6. Controller Logs

Description:
Shows logs for allowed and blocked traffic.

Expected Output:

- "Allowed HTTP (Port 80)"
- "Blocked Telnet (Port 23)"

[Insert Screenshot Here: POX logs]

7. Iperf Test (Optional)

Command:
mininet> iperf h1 h2

Expected Output:
Displays bandwidth between hosts.

[Insert Screenshot Here: Iperf output]

```
mininet> iperf h1 h2
*** Iperf: testing TCP bandwidth between h1 and h2
*** Results: ['25.4 Mbits/sec', '25.1 Mbits/sec']
mininet>
```